

A headline can be numerically true and misleading

Learning intention

Create an accurate disease-data infographic using a checked source, comparable population measures and a cautious conclusion.

“Town C has the most cases.”

Does that tell us which town has the largest proportion affected?

Read what the number means

R1. What does a denominator tell us?

R2. What labels does a bar chart need?

R3. Does a pattern alone prove a cause?

Research the source, not just the number

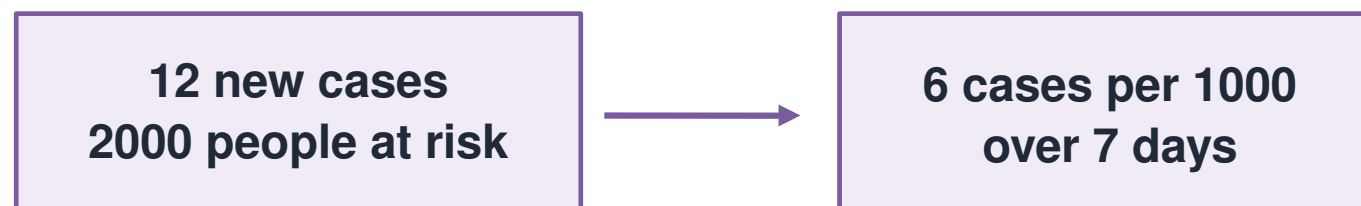
WHO page published 2025; data describe 2021. The statistic measures deaths, not people infected.
Ask: who counted what, where and when?

Source check	What the card says
Publisher	WHO
Published / data	25 Sep 2025 /2021
Measure	Non-communicable disease deaths
75% denominator	Non-pandemic-related global deaths

Real WHO source; figures are estimates for 2021, not today's school data.

Work one fair population comparison

I keep the denominator in the calculation



$$12 \div 2000 \times 1000 = 6$$

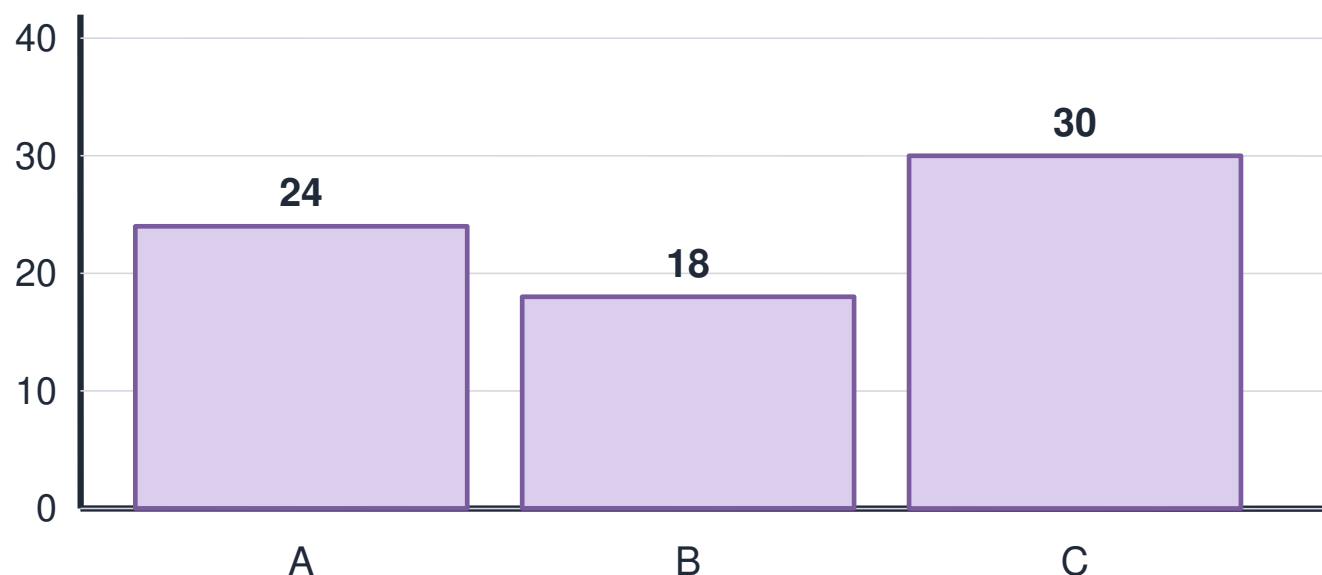
Same disease definition, same period,
comparable recording and population.

I divide by the relevant population first. Then I express the result per 1000.

A count answers one useful question

Constructed town data: count of new cases

Recorded new cases over 7 days



Counts show how many cases were recorded. C has 30; B has 18. Population sizes still differ.

Newsroom calculation desk

W1. Calculate each town's cases per 1000. Keep the same 7-day period in the label.

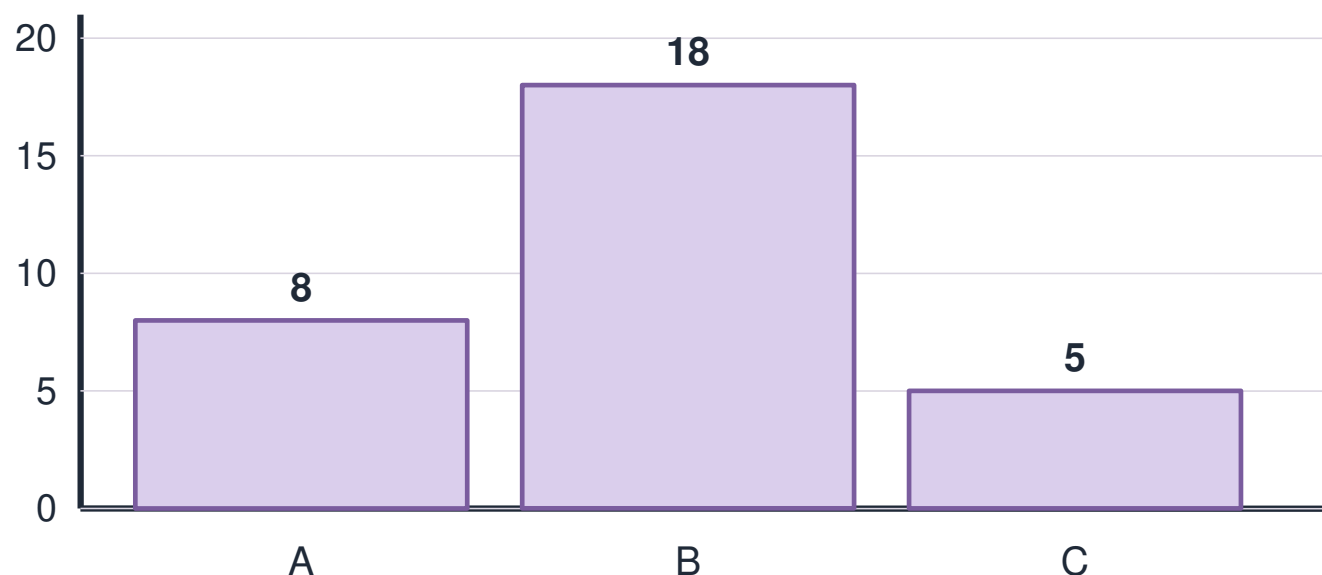
Town	New cases	People at risk
A	24	3000
B	18	1000
C	30	6000

Constructed data: same fictional infection, same 7 days and recording method.

Switch the measure, test the headline

Constructed town data: cases per 1000

Recorded new cases per 1000 people over 7 days



W2. Which town leads by count? Which leads per 1000? Write the two different headlines.

Editors reject an unsupported cause

W3. “B’s water caused the difference.”

The table records no water exposure. Improve the claim.

Which caption is accurate?

Check measure, denominator and period.

- A** C has the largest proportion because it has 30 cases.
- B** B recorded 18 new cases per 1000 people in the same 7 days.
- C** The WHO 75% means 75% of all people died.
- D** A page published 2025 must contain 2025 measurements.

Make a useful infographic

Choose your response route.

Include a fair comparison and a clear source note.

Your one-page newsroom brief

Headline • labelled bars • one calculation.

Source/date • careful conclusion • one limitation.

An editor checks the denominator

L1. Show your infographic to a partner, teacher or TA.

Audience asks: “Per 1000 of whom, during what period?”

Publish only what the evidence supports

A count is not a population comparison.

A chart is not proof of a cause.

Keep the denominator

- E 1. Calculate 10 cases among 2000 per 1000.
- E 2. Why give the data year?
- E 3. Does the town table prove a cause?

Next useful step

Calculate • label • check a source • limit a claim.